crystal for the ordinary ray is 1.658 and for extraordinary ray is 1.486. The refractive index of canada balsam is 1.523.

The ordinary ray is total internally reflected at the layer of canada balsam and is prevented from emerging along with extraordinary ray. Where as, the extraordinary ray is transmitted through the crystal which is plane polarised.

Drawbacks of Nicol prism

(i) Its cost is very high due to scarcity of large and flawless calcite crystals.

(ii) Due to extraordinary ray passing obliquely through it, the emergent ray is always displaced a little to one side.

(iii) The effective field of view is quite limited.

(iv) The light emerging out of it is not uniformly plane polarised.

7.5.8 Polarisation by scattering

When sun light gets scattered by the atmospheric molecules, the electrons of these molecules are influenced by the vibrating components of the electric field present in the sun light. As the sunlight is unpolarised, it produces these vibrations in all directions. These vibrating electrons radiate energy only in the direction perpendicular to their vibrations. When an observer views a beam of sunlight perpendicular to its direction of travel, the radiations produced by the electrons vibrating in the direction perpendicular to the direction of view will only reach the observer. Hence, the light reaching the observer is plane polarised. It is shown in Figure 7.35.

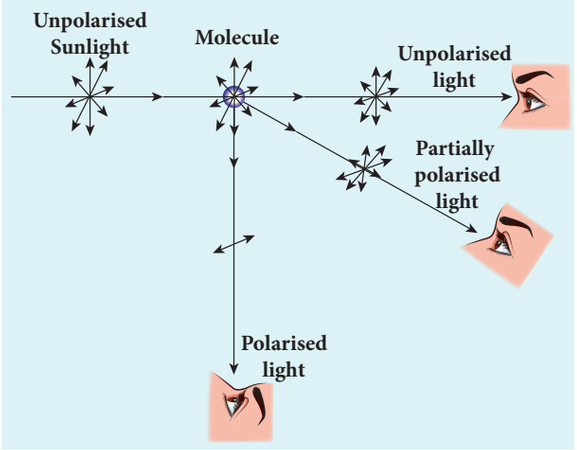


Figure 7.35 Polarisation by scattering

7.6 OPTICAL INSTRUMENTS

There are many optical instruments we used in our daily life. We shall discuss here about microscope, telescope, spectrometer and of course the human eye.

7.6.1 Simple microscope

A simple microscope is a single magnifying (convex) lens of small focal length which must produce an erect, magnified and virtual image of the object. Hence, the object must be placed within the focal length f (between the points F and P) on one side of the lens and viewed through the other side of it. The nearest point where an eye can clearly see is called the near point and the farthest point up to which an eye can clearly see is called the far point. For a healthy eye, the distance of the near point is 25 cm, which is denoted as D and the far point should be at infinity.

7.6.1.1 Near point focusing

The eye is least strained when image is formed at near point, i.e. 25 cm. The near point is also called as *least distance of*

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